# Optimal strategies in the all-heads coin game

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**Abstract**

We study a sequential coin-flipping game in which a player starts with $n$ coins, each landing heads independently with probability $p$. In each round the player flips all remaining coins and must set aside at least one coin showing heads; if no coin shows heads, the player loses. The player wins if and when all coins have been set aside. This is a stochastic shortest-path Markov decision process whose Bellman equation involves a nonlinear suffix-maximum operator. We analyse two natural strategies — One (set aside exactly one head) and All (set aside every head) — and determine the optimal winning probability $w_{n,p}$ as a function of $n$ and $p$.

For $p = \frac{1}{2}$ every strategy that accepts the immediate win achieves $w_{n,1/2} = \frac{1}{2}$. For $p > \frac{1}{2}$ the strategy One is optimal and $n \mapsto w_{n,p}$ is strictly increasing. For $p < \frac{1}{2}$ near $\frac{1}{2}$ we carry out a first-order perturbation expansion in $\delta := \frac{1}{2} - p$ and prove that the leading coefficient $c_n$ of the deficit $\frac{1}{2} - w_{n,p}$ satisfies a linear recursion for $n \geq 7$, with a limit $L = \lim c_n \approx 1.7035$. As a consequence the optimal value sequence has, to first order in $\delta$, a strict local minimum at $n = 5$ and no local maximum — local maxima are a non-perturbative phenomenon.

# 1 Introduction

## 1.1 The game

Fix a probability $p \in (0, 1)$ and a positive integer $n$. A player is given $n$ coins, each of which independently lands heads with probability $p$ and tails with probability $q := 1-p$. The game proceeds in rounds. In each round:

(i) All remaining coins are flipped simultaneously.

(ii) If no coin shows heads, the game is lost.

(iii) Otherwise the player must set aside at least one coin that shows heads. The set-aside coins leave the game permanently.

(iv) If every coin has been set aside, the game is won. Otherwise, a new round begins with the remaining coins.

The player's objective is to maximise the probability of winning.

Two natural strategies present themselves:

- **Strategy One:** set aside exactly one head when $1 \le k \le n-1$ heads appear, and set aside all $n$ when $k = n$.
- **Strategy All:** set aside every head that appears ($i = k$).

Strategy One is conservative: it preserves as many coins as possible for future rounds, betting that the one set-aside coin has already "done its job." Strategy All is greedy: it cashes in every success immediately. Neither is universally optimal.

A variant of this game, in which the objective is to maximise the expected number of heads collected rather than the probability of collecting all of them, was introduced and studied by van Doorn [6]. The all-heads-wins formulation studied here leads to a richer structural picture.

## 1.2 Markov decision processes and the Bellman equation

The game is a finite-state *Markov decision process* (MDP): the state is the number $m$ of coins remaining, the action is the number $i \in \{1, \dots, k\}$ of heads to set aside (where $k$ is the random number of heads), and the transition is $m \mapsto m - i$. There are two absorbing states: *win* ($m = 0$, all coins collected) and *lose* (reached when $k = 0$ in some round). A *policy* (or *strategy*) $\pi$ specifies, for each state $n$ and each observed number $k \in \{1, \dots, n\}$ of heads, the number $i_\pi(n, k) \in \{1, \dots, k\}$ of heads to set aside; we write $v_n^\pi$ for the resulting winning probability starting from state $n$.

Because every policy absorbs with probability at least $p^m + (1-p)^m > 0$ per round, the MDP is a *stochastic shortest-path* (SSP) problem in the sense of Bertsekas and Tsitsiklis [2]: the Bellman operator is a contraction and the optimal value function is its unique fixed point.

**Definition 1.1** (Optimal winning probability)**.** Let $w_{0,p} := 1$. For $n \ge 1$ define

$$w_{n,p} \;:=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \le m \le n-1} w_{m,p}, \tag{1}$$

where $q = 1 - p$ and the empty sum (at $n = 1$) gives $w_{1,p} = p$.

The term $p^n$ accounts for the event that all $n$ coins land heads (an immediate win). Each summand with $j \ge 1$ accounts for the event that exactly $j$ coins land tails; the player then chooses to keep some number $m \in \{j, \dots, n-1\}$ of coins (setting aside $n - m$ heads), and the game continues from state $m$ with optimal value $w_{m,p}$.

The Bellman equation (1) is the standard optimality equation for this SSP–MDP. For general background on Markov decision processes and the Bellman equation we refer to the monographs of Puterman [4], Bertsekas [1], and Stokey, Lucas and Prescott [5].

## 1.3 Strategies One and All

Under strategy One the player always keeps $m = n - 1$ coins (except when $k = n$), so the Bellman recursion collapses to

$$a_{n,p} \;=\; p^n + \big(1 - p^n - (1-p)^n\big)\, a_{n-1,p}, \qquad a_{0,p} := 1. \tag{2}$$

Under strategy ALL the player always keeps $m = j$ coins (the number of tails), giving

$$b_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \, b_{j,p}, \qquad b_{0,p} := 1. \tag{3}$$

Note that the recursions for both $a$ and $b$ are *linear* (no max operator), which makes them analytically more tractable than the Bellman equation (1).

### 1.4 Outline

In Section 2 we prove that at $p = \frac{1}{2}$ the optimal winning probability is $w_{n,1/2} = \frac{1}{2}$ for every $n$, and that in fact every reasonable strategy achieves this value. In Section 3 we show that for $p > \frac{1}{2}$ the strategy ONE is always optimal and $n \mapsto w_{n,p}$ is strictly increasing. Section 4 treats the case $p < \frac{1}{2}$ near $\frac{1}{2}$: we introduce a first-order perturbation in $\delta = \frac{1}{2} - p$, derive a recursion for the leading coefficient $c_n$, prove that it simplifies to a linear recursion for $n \geq 7$, and compute the limit $L = \lim c_n$.

## 2 The fair coin: $p = \frac{1}{2}$

**Theorem 2.1.** *For every $n \geq 1$, $w_{n,1/2} = \frac{1}{2}$. More generally, for any policy $\pi$ satisfying $i_\pi(n,n) = n$ for every $n$ (i.e. the player accepts the immediate win when all coins land heads), the winning probability is $v_n^\pi = \frac{1}{2}$.*

*In particular, both strategies ONE and ALL achieve the optimum: $a_{n,1/2} = b_{n,1/2} = w_{n,1/2} = \frac{1}{2}$.*

*Proof.* We use the binomial identity

$$\sum_{j=1}^{n-1} \binom{n}{j} \left(\frac{1}{2}\right)^n \;=\; 1 - 2^{1-n}. \tag{4}$$

*Step 1.* We show $w_{n,1/2} = \frac{1}{2}$ by strong induction. For $n = 1$: $w_{1,1/2} = \frac{1}{2}$. For $n \geq 2$, suppose $w_{m,1/2} = \frac{1}{2}$ for all $1 \leq m \leq n-1$. Then every suffix-maximum equals $\frac{1}{2}$, so

$$\begin{aligned} w_{n,1/2} &= 2^{-n} + \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} 2^{-n} \\ &= 2^{-n} + \tfrac{1}{2}\big(1 - 2^{1-n}\big) \\ &= 2^{-n} + \tfrac{1}{2} - 2^{-n} \;=\; \tfrac{1}{2}. \end{aligned}$$

*Step 2.* For a general policy $\pi$ with $i_\pi(n,n) = n$, conditioning on the number of heads $K \sim \mathrm{Bin}(n, \frac{1}{2})$ gives

$$v_n^\pi = 2^{-n} \cdot 1 + \sum_{k=1}^{n-1} \binom{n}{k} 2^{-n} \, v^\pi_{n - i_\pi(n,k)}.$$

Since $1 \leq i_\pi(n,k) \leq k \leq n-1$ for $k < n$, the subscript $n - i_\pi(n,k)$ lies in $\{1, \ldots, n-1\}$, and by strong induction each $v^\pi_{n-i_\pi(n,k)} = \frac{1}{2}$. The same calculation as in Step 1 gives $v_n^\pi = \frac{1}{2}$. $\square$

*Remark* 2.2. A policy that refuses the immediate win at $k = n$ (i.e. $i_\pi(n, n) < n$) achieves strictly less than $\frac{1}{2}$: the contribution of the $k = n$ event drops from $2^{-n}$ to $2^{-n} \cdot v^\pi_{n-i_\pi(n,n)} < 2^{-n}$, while all other terms remain at most $\frac{1}{2} \cdot \binom{n}{k} 2^{-n}$, giving $v^\pi_n < \frac{1}{2}$.

# 3 Above the fair coin: $p > \frac{1}{2}$

**Theorem 3.1.** *For $p > \frac{1}{2}$ and all $n \geq 2$:*

(i) $w_{n,p} > w_{n-1,p}$, *i.e. $n \mapsto w_{n,p}$ is strictly increasing.*

(ii) $w_{n-1,p} < \dfrac{p^n}{p^n + q^n}$.

(iii) *Strategy* One *is optimal:* $w_{n,p} = a_{n,p}$.

*Proof.* We prove (i) and (ii) simultaneously by strong induction on $n$.

*Base case* $n = 2$. One computes $w_{1,p} = p$ and $w_{2,p} = p^2 + (1 - p^2 - q^2)p = 3p^2 - 2p^3$. Then $w_{2,p} - w_{1,p} = p(1 - p)(2p - 1) > 0$ since $p > \frac{1}{2}$. Also $w_{1,p} = p < p^2/(p^2 + q^2)$ iff $p^2 + q^2 > p$, which holds for $p \in (\frac{1}{2}, 1)$.

*Inductive step.* Assume (i) and (ii) hold for all indices up to $n - 1$. Since $n \mapsto w_{n,p}$ is increasing on $\{1, \ldots, n - 1\}$ by hypothesis, the suffix-maximum $\max_{j \leq m \leq n-1} w_{m,p} = w_{n-1,p}$ for every $j \in \{1, \ldots, n - 1\}$. Hence the Bellman equation (1) reduces to

$$w_{n,p} = p^n + (1 - p^n - q^n)\, w_{n-1,p},$$

which is exactly the recursion (2) for strategy One. This proves (iii).

For (i), $w_{n,p} > w_{n-1,p}$ iff $p^n(1 - w_{n-1,p}) > q^n w_{n-1,p}$, i.e. $w_{n-1,p} < p^n/(p^n + q^n)$, which is hypothesis (ii) at level $n - 1$.

For (ii) at level $n$, we need $w_{n,p} < p^{n+1}/(p^{n+1} + q^{n+1})$. Using the recursion and the inductive bound on $w_{n-1,p}$, a direct computation shows this reduces to the inequality $(q/p)^n < (q/p)^{n-1}$, which holds since $q/p < 1$. □

# 4 Below the fair coin: perturbation in $\delta = \frac{1}{2} - p$

For $p < \frac{1}{2}$ the structure of $n \mapsto w_{n,p}$ is far richer: numerical experiments show that the sequence develops local minima and maxima, and the optimal strategy is neither One nor All in general; see [6], Section 10.

To get analytical traction, we study the regime $p = \frac{1}{2} - \delta$ with $\delta > 0$ small.

## 4.1 The deficit and its recursion

**Definition 4.1.** For $p \in (0, 1)$ and $n \geq 1$, define the *deficit* $\Delta_{n,p} := \frac{1}{2} - w_{n,p}$.

By Theorem 2.1, $\Delta_{n,1/2} = 0$ for all $n$.

**Proposition 4.2.** *The deficit satisfies*

$$\Delta_{n,p} \;=\; \frac{q^n - p^n}{2} + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \le m \le n-1} \Delta_{m,p}, \tag{5}$$

*with* $\Delta_{0,p} = -\frac{1}{2}$. *In particular:*

(i) *For* $p < \frac{1}{2}$: $\Delta_{n,p} > 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} < \frac{1}{2}$*).*

(ii) *For* $p > \frac{1}{2}$: $\Delta_{n,p} < 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} > \frac{1}{2}$*).*

*Proof.* Substitute $w_{m,p} = \frac{1}{2} - \Delta_{m,p}$ into the Bellman equation (1). The max on $w_m$ becomes a min on $\Delta_m$. The constant terms $\frac{1}{2} \cdot \big[\text{binomial weights}\big]$ cancel by the identity $p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \cdot \frac{1}{2} = \frac{1}{2} - \frac{1}{2}(q^n - p^n)$, leaving (5). The sign claims follow by induction using $q^n - p^n > 0$ (resp. $< 0$) and the positivity (resp. negativity) of the min terms. □

## 4.2 First-order expansion

Set $p = \frac{1}{2} - \delta$ with $\delta > 0$ small. Then $q = \frac{1}{2} + \delta$, and the constant term in (5) is

$$\frac{q^n - p^n}{2} = \frac{(\frac{1}{2} + \delta)^n - (\frac{1}{2} - \delta)^n}{2} = \frac{n\delta}{2^{n-1}} + O(\delta^3).$$

The binomial weights become $\binom{n}{j} 2^{-n} + O(\delta)$, and the min terms are $O(\delta)$ by Proposition 4.2. Thus $\Delta_{n,p} = O(\delta)$.

**Proposition 4.3** (First-order coefficient)**.** *Write* $\Delta_{n,p} = c_n \delta + O(\delta^2)$ *as* $\delta \to 0^+$. *Then* $c_1 = 1$ *and for* $n \ge 2$*:*

$$c_n \;=\; \frac{n}{2^{n-1}} + \frac{1}{2^n} \sum_{j=1}^{n-1} \binom{n}{j} \min_{j \le m \le n-1} c_m. \tag{6}$$

*Proof.* Match first-order coefficients in (5). The constant term contributes $n/2^{n-1}$. Each binomial weight is $\binom{n}{j} 2^{-n} + O(\delta)$; the min of the $\Delta_m = c_m \delta + O(\delta^2)$ at first order is $(\min c_m)\delta + O(\delta^2)$. Products of $O(\delta)$ terms are $O(\delta^2)$. □

**Example 4.4.** The first few values of $c_n$:

$$c_1 = 1, \quad c_2 = \tfrac{3}{2}, \quad c_3 = \tfrac{27}{16}, \quad c_4 = \tfrac{111}{64}, \quad c_5 = \tfrac{3555}{2048} \approx 1.7358.$$

The sequence increases on $\{1, \ldots, 5\}$ and decreases for $n \ge 6$.

## 4.3 Collapse of the minimum

The key structural insight is that for $n \ge 7$, the min in (6) can be evaluated explicitly.

**Lemma 4.5** (Asymptotic collapse)**.** *For all* $n \ge 7$ *and* $j \in \{1, \ldots, n-1\}$*:*

$$\min_{j \le m \le n-1} c_m \;=\; \begin{cases} c_j & \text{if } j \in \{1, 2, 3\}, \\ c_{n-1} & \text{if } j \in \{4, \ldots, n-1\}. \end{cases} \tag{7}$$

The proof of Lemma 4.5 requires two auxiliary claims.

**Lemma 4.6.** *$c_m \geq \frac{27}{16}$ for every $m \geq 4$.*

*Proof.* We use the algebraic identity

$$A_n - \tfrac{27}{16} B_n \;=\; -\frac{3(n^2 - 15n + 36)}{32 \cdot 2^n}, \tag{8}$$

where $A_n$ and $B_n$ are defined in (10) below.

*For $m \in \{4, \ldots, 12\}$:* the quadratic $n^2 - 15n + 36$ has roots at $n = 3$ and $n = 12$, so it is non-positive on $\{3, \ldots, 12\}$. Hence $A_m - \frac{27}{16} B_m \geq 0$ for these values, and the linear recursion (9) gives $c_m \geq \frac{27}{16}$ by induction.

*For $m \geq 13$:* define the deficit $\varepsilon_m := c_m - \frac{27}{16}$. From the linear recursion,

$$\varepsilon_m \;=\; (1 - B_m)\varepsilon_{m-1} + \big(A_m - \tfrac{27}{16} B_m\big).$$

The starting value $\varepsilon_{12} = c_{12} - \frac{27}{16} > \frac{1}{60}$ (verified by exact rational computation). The cumulative negative contribution is bounded by $\sum_{k \geq 13} \frac{3(k^2 - 15k + 36)}{32 \cdot 2^k} \leq \frac{3}{32} \cdot \frac{99}{2048} = \frac{297}{65536} < \frac{1}{200}$, using $\sum_{k \geq 13} k^2/2^k = \frac{99}{2048}$. Meanwhile $\prod_{m \geq 13}(1 - B_m) > \frac{7}{8}$ (by $\sum_{m \geq 13} B_m = \frac{243}{2048} < \frac{1}{8}$). So $\varepsilon_m > \frac{1}{60} \cdot \frac{7}{8} - \frac{1}{200} = \frac{23}{2400} > 0$ for all $m \geq 13$. □

**Lemma 4.7.** *The sequence $(c_n)_{n \geq 5}$ is strictly decreasing.*

*Proof.* For $n \geq 7$, the linear recursion gives $c_n - c_{n-1} = A_n - B_n c_{n-1}$. By Lemma 4.6, $c_{n-1} \geq \frac{27}{16}$, so $c_n - c_{n-1} \leq A_n - \frac{27}{16} B_n < 0$ for $n \geq 13$ (by (8) with $n^2 - 15n + 36 > 0$). For $n \in \{5, \ldots, 12\}$ one checks $c_n < c_{n-1}$ by direct computation. □

*Proof of Lemma 4.5.* For $j \in \{1, 2, 3\}$: since $c_1 < c_2 < c_3 < \frac{27}{16} \leq c_m$ for $m \geq 4$ (Lemma 4.6), the minimum over $\{j, \ldots, n-1\}$ is $c_j$.

For $j \geq 4$: all values $c_j, \ldots, c_{n-1}$ are at least $\frac{27}{16}$, and the sequence is decreasing for indices $\geq 5$ (Lemma 4.7). For $j = 4$ one checks $c_4 > \min(c_5, \ldots, c_{n-1}) = c_{n-1}$ since $c_4 = \frac{111}{64} \approx 1.734 > c_6 \approx 1.729$. Hence the minimum is $c_{n-1}$. □

## 4.4 Linear recursion and the limit

**Proposition 4.8** (Linear recursion)**.** *For $n \geq 7$:*

$$c_n \;=\; A_n + (1 - B_n)\, c_{n-1}, \tag{9}$$

*where*

$$A_n := \frac{n}{2^{n-1}} + \frac{n\, c_1 + \binom{n}{2} c_2 + \binom{n}{3} c_3}{2^n}, \qquad B_n := \frac{2 + n + \binom{n}{2} + \binom{n}{3}}{2^n}. \tag{10}$$

*Proof.* Substitute (7) into (6) and split the sum at $j = 3$. For $j \in \{1, 2, 3\}$ the min is $c_j$; for $j \geq 4$ the min is $c_{n-1}$. Collecting terms gives (9)–(10). □

**Theorem 4.9** (Existence and value of the limit)**.** *The limit* $L := \lim_{n\to\infty} c_n$ *exists, and for any* $n_0 \geq 7$*:*

$$L \;=\; c_{n_0-1} \prod_{m\geq n_0} (1 - B_m) \;+\; \sum_{k\geq n_0} A_k \prod_{m>k} (1 - B_m). \tag{11}$$

*Both the infinite product and the series converge at geometric rate since* $A_n, B_n = O(n^3/2^n)$*. Numerically,*

$$L \;=\; 1.70347176087173673645\ldots \tag{12}$$

*Proof.* From (9), $c_n = A_n + (1 - B_n)c_{n-1}$. Since $B_n > 0$ and $A_n > 0$ for all $n$, and $(c_n)$ is eventually decreasing and bounded below (by Lemma 4.6), the limit exists. Iterating the recursion from $n = n_0$ and taking $n \to \infty$ yields (11); the rate follows from $B_n = O(n^3/2^n)$. □

### 4.5 Shape of the optimal-value sequence near $p = \frac{1}{2}$

**Corollary 4.10.** *For* $\delta > 0$ *sufficiently small and* $p = \frac{1}{2} - \delta$*:*

(i) $w_{n-1,p} - w_{n,p} = (c_n - c_{n-1})\delta + O(\delta^2)$.

(ii) *The sequence* $n \mapsto w_{n,p}$ *has, to first order in* $\delta$*, a strict local minimum at* $n = 5$ *(since* $c_5 > c_4$ *and* $c_5 > c_6$*).*

(iii) *There is no local maximum at first order:* $c_n$ *is eventually decreasing, so* $w_{n,p}$ *is eventually increasing.*

*Remark* 4.11. Numerical computation shows that the sequence $n \mapsto w_{n,p}$ does develop strict local maxima for $p$ bounded away from $\frac{1}{2}$ (e.g. at $n = 9$ for $p = 0.42$, verified by exact rational arithmetic). This is consistent with the corollary: local maxima are a *non-perturbative* phenomenon invisible to the first-order expansion. Van Doorn [6], Section 10, conjectures that the optimal strategy for $p < \frac{1}{2}$ is "infinitely complex."

## References


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